

Experimental validation and enhanced thermal prediction of a shallow solar pond using artificial neural network–based model predictive control for real-time optimization under multiple heat extraction modes

Abdelkrim Terfai^a, Younes Chiba^{a,*}, Mounir Zirari^a, Mohamed Najib Bouaziz^b

^a Renewable Energy and Materials Laboratory (LERM), Mechanical Engineering Department, University of Medea, 26000, Algeria

^b Biomaterials and Transport Phenomena Laboratory (LBMP), Mechanical Engineering Department, University of Medea, 26000, Algeria

ARTICLE INFO

Keywords:

Shallow solar pond
Artificial neural networks
Model predictive control
Heat extraction modes
Thermal performance prediction
Solar energy systems
Optimization algorithms

ABSTRACT

This study presents the experimental validation and enhanced thermal modeling of a Shallow Solar Pond (SSP) operating under three distinct heat extraction modes: direct, open cycle, and closed cycle. A custom-designed SSP, insulated and equipped with double-glazing and a PVC heat exchanger and black-painted bottom, was monitored under clear-sky conditions. 14 Artificial Neural Network (ANN) configurations were trained, with the optimal Bayesian Regularization-based model (two hidden layers, 15 neurons) achieving exceptional predictive accuracy ($R^2 = 0.99919$, $RMSE = 32.7580$ %, $MRE = 0.62$ %) for temperatures across all system components and operational modes. The ANN model effectively captured complex nonlinear dynamics, including transient heating and cooling phases. The integration of this ANN with a Model Predictive Control (MPC) framework enabled real-time optimization of the fluid flow rate in the closed-cycle mode, which was identified as the most efficient configuration. The MPC-ANN strategy reduced the outlet temperature tracking error by over 50 % ($MAE = 1.42$ °C vs. 2.97 °C without MPC), demonstrating superior thermal stability and adaptive control under fluctuating solar irradiance. This work confirms the closed-cycle SSP's superiority for stable energy delivery and establishes a robust ANN-MPC framework for real-time operational optimization of solar thermal systems.

1. Introduction

The development of renewable energy technologies is a global priority due to the growing demand for sustainable energy solutions. One of the most promising alternatives is solar energy for its abundance and environmentally friendly nature [1]. Among various solar energy technologies, the shallow solar pond (SSP) has attracted significant attention due to its simplicity, cost-effectiveness, and the ability to store thermal energy and use it for various applications [2]. An SSP typically consists of a shallow water body with a blackened bottom designed to absorb and retain solar radiation, converting it into heat [3]. This stored heat can be harnessed for tasks such as water heating, industrial processes, and even power generation. Several design enhancements, including the use of transparent covers, thermal insulation, and integrated heat exchangers, have been investigated to boost system efficiency [4]. Heat extraction from SSPs is categorized into three primary methods: the direct method, which involves draining the pond to access heated water, albeit with

lower efficiency; the open-cycle mode, where a heat exchanger circulates fluid through the pond; and the closed-cycle mode, which employs a sealed heat exchanger system to reduce thermal losses and optimize heat transfer [4–6].

Significant research advancements have refined SSP technology. Early work by Dickinson et al. [3] demonstrated the feasibility of large-scale power generation using SSPs, emphasizing transparent plastic layers to minimize heat loss. Further studies [4,5] underscored the importance of optimizing pond depth, glazing materials, and operational modes, achieving efficiencies of up to 50 %. More recently, the introduction of nanofluids, such as aluminum oxide (Al_2O_3) and copper oxide (CuO), has been explored to enhance heat transfer efficiency [7]. To further improve thermal storage capabilities and reduce energy losses, researchers have incorporated serpentine heat exchangers, yielding efficiencies up to 60 % under specific conditions [6]. Experimental work by Ali et al. [8] validated the influence of water depth and glazing configurations on performance, while [9] confirmed the effectiveness of

Peer review under the responsibility of KeAi Communications Co., Ltd.

* Corresponding author.

E-mail address: chiba.younes@univ-medea.dz (Y. Chiba).

<https://doi.org/10.1016/j.unres.2025.100240>

Received 25 March 2025; Received in revised form 2 September 2025; Accepted 4 September 2025

Available online 5 September 2025

2666-5190/© 2025 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

dark-colored water bags for tropical environments. Studies by Velmuruganand Srithar [10], El-Sebaï et al. [11] and El-Sebaï et al. [12] demonstrated the successful integration of SSPs with solar stills, enhancing productivity.

As the design of SSP is also progressing, Artificial Neural Networks (ANNs) are turning out to be indispensable in the simulation, optimization, and modeling of renewable energy systems. ANNs are artificial intelligence algorithms that emulate the working of the human brain to learn from data and recognize patterns in data sets, using interconnected nodes. Because of their flexibility, they are very useful in solving nonlinear problems in energy studies [13]. Various types of ANNs, including multilayer perceptron (MLP) and radial basis function (RBF) networks, have been widely used in solar energy applications. MLP models, known for their simple architecture and efficient training, have been applied to forecast solar radiation, optimize energy efficiency, and predict thermal performance [14]. ANN models have demonstrated high accuracy in estimating heat transfer rates for parabolic trough, evacuated tube, and solar air collectors [14–16]. Comprehensive reviews further confirm the effectiveness of ANN in predicting thermal performance across various solar thermal systems [17]. In addition, ANN-based approaches have been applied to evaluate the impact of nanofluids, such as Cu_2O and SiO_2 , on solar system efficiency, providing insights into their potential to enhance thermal performance [18,19].

The combination of ANN models with renewable energy research has contributed to improved design and operational strategies for solar thermal systems, including photovoltaic/thermal configurations, solar dryers, and solar-assisted heat pumps [20].

The increasing use of ANNs in renewable energy systems highlights their importance in enhancing sustainable energy performance. In this context, the present study aims to develop, validate, and optimize an ANN-based model for predicting the thermal behavior of an SSP operating under three different heat extraction modes: direct, open, and closed cycles. Experimental data were collected from a custom-built 1 m^2 SSP operating under identical environmental conditions for comparative analysis. The developed ANN model was trained not only to predict thermal performance with high accuracy but was also seamlessly integrated into a real-time Model Predictive Control (MPC) system. This integration enabled intelligent, adaptive regulation of key operating parameters most notably, the fluid flow rate to dynamically optimize heat extraction efficiency under varying solar conditions.

The novelty of this research is threefold. First, it presents a rigorous comparative experimental analysis of three distinct heat extraction modes direct, open cycle, and closed cycle conducted under identical environmental conditions, a comprehensive approach seldom achieved in existing literature. Second, it advances beyond conventional predictive modeling by embedding the ANN as a core component within a real-time control framework, facilitating online thermal optimization. Third, the SSP was specially designed, constructed, and instrumented to enable high-resolution, real-time thermal monitoring across multiple points, significantly enhancing the reliability and fidelity of both model training and experimental validation.

The paper is organized as follows: Section 1 outlines the background, motivation, and research objectives; Section 2 details the experimental setup and data acquisition methodology; Section 3 provides an overview of ANN fundamentals and training techniques; Section 4 specifies the modeling parameters and performance evaluation metrics; Section 5 compares ANN predictions with experimental results; Section 6 analyzes the thermal performance of each operating mode; Section 7 introduces the real-time control strategy via ANN-MPC integration; and finally, Section 8 summarizes the key findings and contributions.

2. Construction of the system and data collection

Fig. 1 illustrates the experimental setup of the SSP. The system was constructed using galvanized sheet materials with dimensions of 0.76×1.30 m and a depth of 0.06 m, accommodating a water capacity of

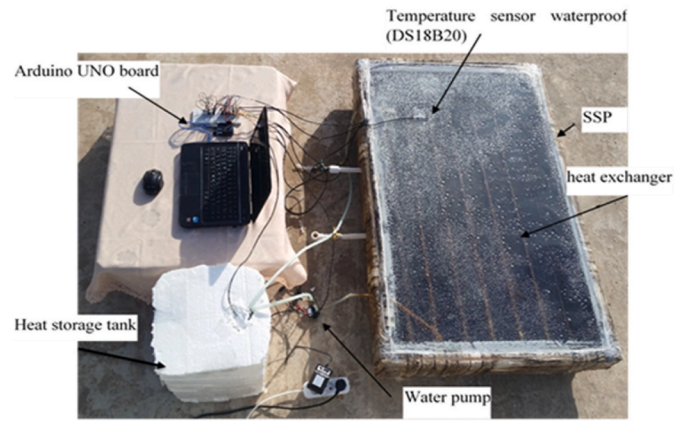


Fig. 1. Photograph of the experimental setup used for SSP.

approximately 60 L. To minimize conductive heat losses, the pond was thermally insulated with a 0.04 m polystyrene layer, while the bottom surface was painted black to maximize solar radiation absorption. Heat extraction was achieved through a transparent PVC heat exchanger, 10 m in length and 0.008 m in diameter, connected to a QR30E brushless pump and a 20-L insulated storage tank. Although the effectiveness (ϵ) of the heat exchanger was not directly measured in this work, its thermal performance can be reasonably inferred from the material properties. Considering the thermal conductivity of PVC ($\lambda \approx 0.19$ W/m·K) and the tube wall thickness ($\delta = 0.001$ m), the overall heat transfer coefficient (U) is expected to be relatively low compared to metallic exchangers. However, the experimental results showed a stable temperature rise and consistent heat transfer, indicating that the exchanger operated with satisfactory effectiveness. Tap water was employed as the working fluid due to its availability, cost-effectiveness, and favorable thermal properties. Its main thermophysical characteristics are summarized in Table 1, including thermal conductivity ($k = 0.6$ W/m·K), specific heat capacity ($C_p = 4180$ J/kg·K), and density ($\rho = 1000$ kg/m³). These properties justify its selection as a working fluid, as they ensure efficient heat transfer and reliable operation at low cost compared to alternative engineered fluids. The SSP was covered with two glass panels, each 0.003 m thick, separated by a 0.03 m air gap. The use of double glazing serves multiple purposes: (i) reducing convective heat losses through the insulating air gap, (ii) enhancing the greenhouse effect by transmitting short-wave solar radiation while restricting long-wave thermal emission, and (iii) protecting the pond against condensation, dust, and environmental effects. As noted by Kamiuto et al. [21], the application of multilayer transparent insulation significantly improves the thermal performance of solar pond systems by mitigating upward heat losses.

It should be noted that the three experiments were conducted on different consecutive days (August 15–17), under clear-sky conditions without significant weather fluctuations. This ensured that variations in solar radiation or ambient temperature did not bias the comparison across modes. Details of the main materials and components used in constructing the SSP are provided in Table 1, which complements the schematic illustrated in Fig. 1.

2.1. Experimental scenarios

To investigate the performance of the SSP under different thermal extraction strategies, three experimental scenarios were designed. The rationale for selecting direct, open, and closed cycles lies in their distinct heat transfer characteristics and their relevance to practical solar thermal applications. The direct mode represents the simplest case, the open cycle reflects intermediate efficiency with continuous heat removal, while the closed cycle provides maximum isolation and storage capacity. These scenarios were therefore chosen to cover the full spectrum of SSP operation strategies reported in the literature.

Table 1

Summary of materials and components used in the SSP experimental setup.

Component	Material/Type	Dimensions			Numerical value		
Transparent cover	Double glass panels						
		thickness		air gap	α_g	τ_g	ε_g
		0.003 m		0.03 m	0.05	0.90	0.90
Working fluid	Tap water	–			K_w 0.6 W/m·K	C_w 4180 J/kg·K	ρ_w 1000 kg/m ³
Heat exchanger	PVC tube	length 10 m	diameter 0.008 m	thickness 0.001 m	K_{PVC} 0.19 W/m·K		
Thermal insulation	Polystyrene	thickness 0.04 m			$K_{Polystyrene}$ 0.03 W/m·K		
Absorber plate	galvanized sheet	length 1.3 m	width 0.76 m		α_p 0.95		

The experiments were not operated simultaneously, but separately, one per day. This approach was necessary to avoid thermal interference between modes and to ensure independent performance evaluation of each configuration:

1. Direct mode heat extraction (August 15)

In the first scenario, thermal energy was extracted using the direct mode, the simplest approach among the three (see Fig. 2). Heated water was drained directly from the pond, allowing immediate use of the stored heat. While this method enabled quick access to thermal energy, it resulted in significant heat losses due to exposure and the lack of thermal insulation or recovery mechanisms.

2. Open cycle heat extraction (August 16)

The second scenario implemented an open-cycle continuous flow system. In this configuration, a heat exchanger placed inside the pond (illustrated in Fig. 3) transferred heat to a circulating working fluid, which continuously removed thermal energy from the pond. This setup enabled steady and sustained heat transfer, making it more suitable for applications that require a continuous thermal supply. However, being an open cycle, some thermal losses still occurred.

3. Closed cycle heat extraction with thermal storage tank (August 17)

In the third scenario, the system operated in a closed-cycle continuous flow heating mode, where the working fluid circulated in a sealed loop between the SSP and an insulated storage tank. A serpentine heat exchanger was placed inside the pond to collect thermal energy, and a second heat exchanger was installed inside the thermal storage tank to transfer the collected heat as shown in Fig. 4. This design effectively isolated the working fluid from direct contact with the pond water, significantly reducing heat losses. As a result, it offered higher thermal efficiency and longer heat retention, making it more suitable for applications requiring stable and delayed thermal delivery. This design effectively minimized contact with the environment, thus reducing heat losses and enhancing long-term heat storage, making it suitable for

delayed thermal utilization.

2.2. Data collection and analysis

Data acquisition was performed using an Arduino UNO microcontroller, complemented by various sensors, with measurements recorded every minute from 07:00 to 19:00 h. This experimental procedure was strictly followed for each of the three scenarios to maintain consistency and comparability across modes. The ambient air temperature and humidity were monitored using a DHT22 sensor, while seven DS18B20 waterproof temperature probes were installed at critical locations, including the upper and lower glass covers, the storage tank, the absorber plate, the pond water, and the inlet/outlet of the heat exchanger. The QR30E brushless pump was employed to circulate the heat transfer fluid and overcome the pressure drop. Solar radiation was estimated using the semi-empirical Bird & Hulstrom model [22], based on the geographical coordinates of Tablat, Medea, Algeria.

The accuracy and uncertainty of the employed sensors are summarized in Table 2, which ensures transparency in evaluating the reliability of the experimental results. The DHT22 sensor features a temperature accuracy of ± 0.5 °C and a humidity accuracy of ± 2 % RH, while the DS18B20 sensors provide a measurement precision of ± 0.5 °C within the range of -10 °C to 85 °C. Such accuracies are consistent with the requirements for thermal performance assessment of solar pond systems and guarantee reliable model validation against experimental data. The uncertainty analysis (Table 2) ensures the reliability of the obtained results and strengthens confidence in the subsequent ANN modeling and performance evaluation.

3. Overview of ANNs

ANNs are an essential part of AI along with expert systems, fuzzy logic, and automatic control systems. ANNs provide a scientific approach to developing models that learn from data, identify patterns, and make data-driven decisions. The bio-inspired architecture of ANNs mimics the parallel processing capability of the human brain by utilizing interconnected processing units known as neurons [13,14].

3.1. Key features of ANNs

The effectiveness of ANNs lies in their ability to handle complex and imprecise data. The most significant features of ANNs include:

- **Pattern Recognition and Learning:** ANNs can extract useful information from noisy or complex data sets, making pattern recognition and classification of data possible.
- **Adaptive Learning:** ANNs adjust their internal parameters (weights) by training or experiences to enhance performance, learning to carry out data-dependent tasks.

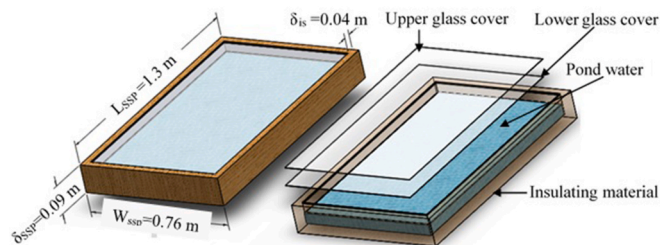


Fig. 2. Schematic representation of the Direct Heat Extraction Mode.

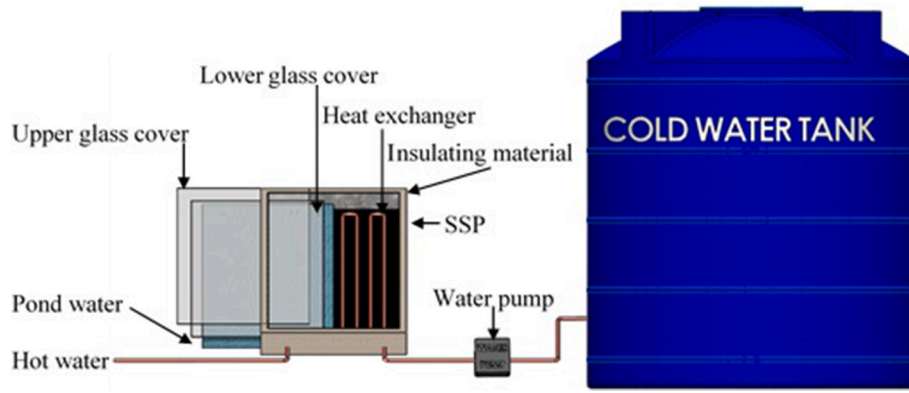


Fig. 3. Thermal operation layout of an SSP in open-cycle continuous flow mode.

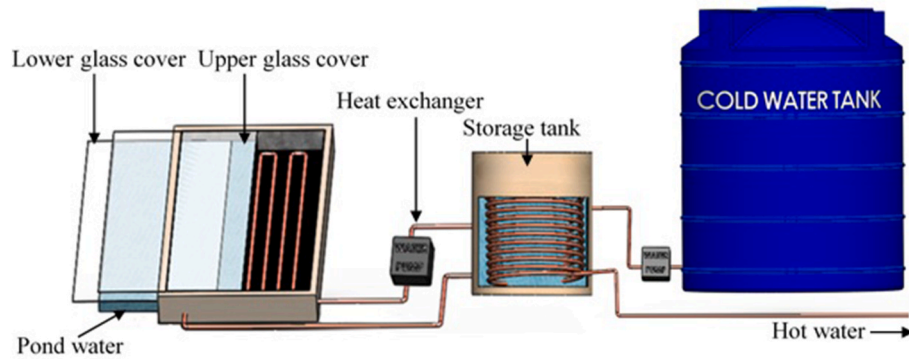


Fig. 4. Schematic diagram of an SSP operating in closed-cycle continuous flow heating mode.

Table 2

Specifications and accuracy of the measurement devices.

Device/ Sensor	Parameter Measured	Measuring Range	Accuracy/ Error	Notes/ Application
DHT22 sensor	Ambient air temperature	−40 to +80 °C	±0.5 °C	Used for monitoring ambient conditions
DHT22 sensor	Relative humidity	0–100 % RH	±2 % RH	Connected to Arduino via serial port
DS18B20 waterproof probe	Pond, glass, absorber, tank, exchanger temperatures	−55 to +125 °C	±0.5 °C (−10 to +85 °C)	Installed at 7 key thermal nodes
Arduino Uno	Data acquisition/ control	–	–	Logged sensor readings every minute
QR30E brushless pump	Fluid circulation	Max flow 240 L/h	–	Ensures heat transfer and stable flow

- Self-Organization: ANNs self-organize their internal representation of data during learning, offering dynamic modeling of data.
- Parallel Processing: ANNs distributed structure supports simultaneous computations, enhancing processing efficiency and robustness [23].

3.2. ANN architecture and components

The architecture of ANNs consists of layers and neurons that are interconnected and each contributes to the overall functionality of the network. The most important components are:

3.2.1. Network layers

- Input Layer: Receives input features such as temperature, humidity, and solar radiation.
- Hidden Layers: Consist of artificial neurons that process input data using mathematical transformations, allowing the network to learn complex, non-linear relationships [18].
- Output Layer: Produces the ultimate output from calculations of the earlier layers, which is the network classification or prediction.

3.2.2. Artificial neurons and activation functions

Each neuron processes input data using weighted connections, a summation function, and an activation function. Common activation functions include:

- Sigmoid Function: Smooth gradients are generated, which is appropriate for probabilistic outputs.
- Linear Function: Employed where regression is needed.
- Sign Function: Generates binary outputs, useful in classification.
- Step Function: Triggers neurons only when inputs surpass a certain threshold [23].

3.2.3. ANNs types

ANNs may be grouped based on their layer structure:

- Single-layer Neural Networks: Consist of one processing layer, which feeds inputs straight into the output layer without any hidden layers, suitable for simple problems.
- Multi-layer Neural Networks: Are made up of more than one hidden layer, which raises the capacity of the network to model complicated relationships at the cost of greater computational complexity.

4. ANN training and optimization

To ensure the accurate thermal performance prediction of the SSP, different ANN training scenarios were considered. Training was focused on the optimization of the network performance by changing key hyperparameters, which consist of the training algorithm, the number of hidden layers, and the number of neurons in each layer.

4.1. Training algorithm selection

Various backpropagation-based training functions were attempted, including Bayesian Regularization (trainbr), Levenberg-Marquardt (trainlm), BFGS Quasi-Newton (trainbfg), and Scaled Conjugate Gradient (trainscg). All the algorithms were benchmarked against each other based on their error minimization and computational efficiency.

4.2. Data preprocessing and outlier treatment

Prior to training the ANN model, a data preprocessing phase was conducted to ensure the reliability of the dataset. Input variables including temperature, humidity, and solar radiation were normalized to enhance training convergence and reduce computational bias. To identify potential outliers, a combination of visual inspection and statistical criteria specifically the Z-score method—was applied. Data points deviating more than ± 3 standard deviations from the mean were considered potential outliers. However, analysis of the experimental data revealed no significant outliers that warranted exclusion. Therefore, the entire dataset, consisting of approximately 2160 data points collected over three consecutive days (with a 1-min sampling interval from 07:00 to 19:00), was retained for model training, validation, and testing. The dataset was randomly divided into 70 % for training, 15 % for validation, and 15 % for testing to ensure the generalization capability of the ANN model (see Fig. 4).

The input and output variables used in the ANN model, as illustrated in Fig. 5, were carefully selected to reflect the thermal dynamics of the SSP under all three heat extraction scenarios: direct mode, open-cycle mode, and closed-cycle mode. For all scenarios, the common input parameters included ambient air temperature (T_a), humidity (Hum), solar radiation (I_T), time of day (Time), SSP area (A), SSP depth, insulation thickness, and wind speed. In addition to the above, mode-specific input parameters were incorporated, such as the inlet temperature of the heat exchanger (T_{fi}) and the mass flow rate of the fluid ($\dot{m}$) for the open and closed cycles. The output parameters varied depending on the scenario but primarily aimed to reflect key thermal behavior, including upper and lower glass cover temperatures (T_{C1} , T_{C2}), pond water temperature (T_{wp}), absorber plate temperature (T_p), heat exchanger fluid outlet temperature (T_{fo}), and storage tank temperature (T_{wt}). Although the

number of output parameters may appear large, they were intentionally included to ensure the ANN was trained with sufficient information to achieve high predictive accuracy and to effectively capture the complex nonlinear thermal interactions within the SSP.

5. Performance evaluation metrics

The network's performance was evaluated using four standard statistical indicators: Sum of Squared Errors (SSE), Root Mean Square Error (RMSE), Mean Relative Error (MRE), and the Coefficient of Determination (R^2). These metrics are widely adopted in ANN performance evaluation for regression tasks, particularly in thermal systems modeling, and have been used in several previous studies [16,17].

Equations (1)–(4) are utilized to compute SSE, RMSE, R^2 , and MRE, respectively, for evaluation purposes.

Eq (1) can be used to compute the SSE:

$$SSE = \sum_{i=1}^n (e_i - p_i)^2 \quad (1)$$

The number of output data is n , the experimental value is e_i , and the predicted value is p_i .

Eq (2) is used to determine the RMSE:

$$RMSE(\%) = \sqrt{\left(\frac{SSE}{n}\right)} \times 100 \quad (2)$$

Additionally, Eq (3) is used to obtain the R^2 :

$$R^2 = 1 - \frac{SSE}{\sum_{i=1}^n p_i^2} \quad (3)$$

Eq (4) is used to calculate MRE:

$$MRE(\%) = \frac{1}{n} \sum_{i=1}^n \left(\frac{|e_i - p_i|}{e_i} \times 100 \right) \quad (4)$$

The best scenario was selected based on the lowest error values and the highest R^2 score.

Through these training scenarios, the optimal ANN model was determined for accurately predicting the thermal behavior of the SSP, ensuring both precision and computational efficiency.

6. Results and discussion

The procedure followed for conduction of experiments and ANN training consisted of two main stages. First, the experimental campaign was carried out over three consecutive clear-sky days, during which

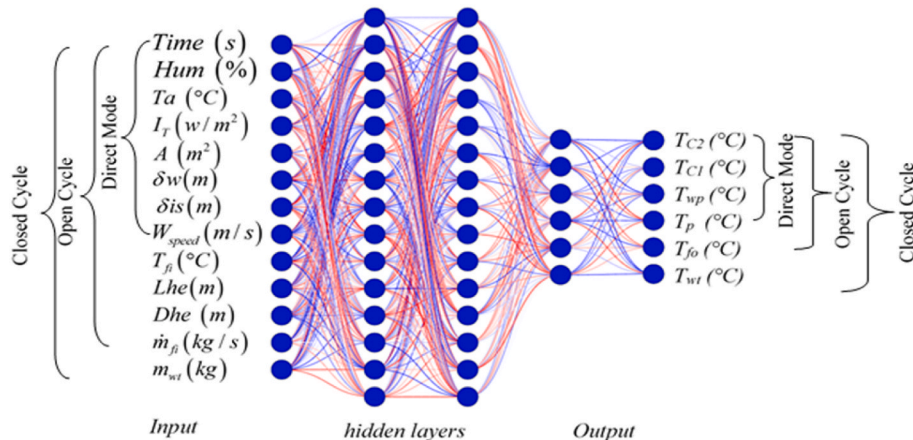


Fig. 5. Schematic diagram of the ANN used in this study.

thermal and environmental parameters were continuously monitored using calibrated sensors connected to an Arduino-based acquisition system. Each heat extraction mode (direct, open, and closed cycles) was tested separately to ensure independent evaluation and comparability. Second, the complete dataset of approximately 2160 measurements per scenario was preprocessed through normalization and outlier detection, then divided into training, validation, and testing subsets. Multiple ANN models were trained using different algorithms, hidden layers, and neuron configurations, and their performance was systematically compared based on error metrics (SSE, RMSE, MRE, and R^2) to identify the optimal predictive model. Based on this procedure, the ANN development stage involved a systematic generation and evaluation of multiple candidate models, as detailed below.

In the ANN development stage, a total of 14 models were generated by systematically varying the training algorithms, the number of hidden layers (1–3), and the number of neurons per layer (5–20). Various backpropagation-based algorithms—including Bayesian Regularization (trainbr), Levenberg-Marquardt (trainlm), BFGS Quasi-Newton (trainbfg), and Scaled Conjugate Gradient (trainscg)—were benchmarked against one another. Each model was rigorously evaluated using statistical performance indicators such as SSE, RMSE, MRE, R^2 , training time, and best epoch. The comparison of these 14 ANN configurations and their corresponding statistical results is summarized in Table 3.

The final ANN model was selected based on its superior performance, particularly the lowest RMSE and the highest R^2 , which demonstrated its capability to capture the non-linear thermal dynamics of the SSP under all operating scenarios.

Fig. 6 presents the correlation between experimental data and ANN-predicted outputs during the training phase of the neural network model. The scatter points (circles) represent the paired values of experimental and predicted temperatures, while the solid line indicates the ideal 1:1 correlation. The coefficient of determination ($R^2 = 0.99919$) demonstrates an almost perfect linear agreement, as the majority of points cluster tightly around the reference line. This close alignment with minimal deviation confirms that the ANN model has successfully captured and internalized the nonlinear thermal dynamics of the shallow solar pond. The negligible spread and offset of the regression line further validate the robustness and accuracy of the model, ensuring reliable generalization for subsequent prediction and control tasks.

Fig. 7 presents a comparative analysis between experimental measurements and ANN-predicted temperatures of the upper glass cover (T_{C2}) under three distinct heat extraction modes: open, closed, and direct. Across all modes, the ANN model demonstrates a high level of accuracy, with maximum deviations remaining below 0.5 °C and MRE values between 0.84 % and 0.91 %. This strong agreement validates the robustness of the developed neural network in capturing the complex thermal dynamics of the shallow solar pond under varying operating

conditions.

In the open cycle, T_{C2} starts at 33.74 °C at 07:00 h and increases steadily to a maximum of 48.63 °C between 13:00 and 14:00 h.

The ANN predictions closely track this rise, with only minor discrepancies at peak values. The temperature increase is primarily attributed to the progressive growth of incident solar radiation and the greenhouse effect induced by the double-glass cover, which traps longwave radiation from the pond surface. The relatively high peak reflects strong convective coupling between the circulating fluid and the pond, enhancing heat extraction while still allowing significant solar energy gain.

For the closed cycle, temperatures begin at 25.00 °C and rise to a peak of 47.94 °C around midday (MRE = 0.91 %). Although starting values are lower than in the open cycle due to reduced convective losses in early morning, the closed configuration shows effective heat retention by minimizing evaporative and convective losses to the surroundings. The ANN model again mirrors the experimental data with high precision, though slight deviations appear during peak hours. These are likely linked to transient fluctuations in solar irradiance and convective exchanges with ambient air, which are inherently difficult to capture simultaneously by sensors and predictive models. Overall, this mode achieves a more balanced performance, providing significant heat extraction while maintaining higher stability compared to the open configuration.

In the direct extraction mode, T_{C2} increases from 25.56 °C in the morning to a maximum of 42.63 °C at noon, which is significantly lower than the peaks observed in the open and closed modes. ANN predictions reproduce this trend accurately, with minor mismatches (MRE = 0.89 %) during the post-peak cooling phase.

The reduced maximum temperature reflects the weaker thermal retention capacity of this mode, as direct extraction intensifies convective losses and reduces the residence time of heated water within the pond. The faster decline in the afternoon is mainly due to enhanced radiative losses from the glass cover to the cooler sky and convective exchanges with ambient air.

Fig. 8 depicts a comparative evaluation between experimental data and ANN-predicted values of the lower glass cover temperature (T_{C1}) under the three heat extraction configurations: open, closed, and direct. Across all modes, the ANN model exhibits excellent agreement with the measured data, with maximum deviations below 0.5 °C and MRE values ranging from 0.42 % to 0.48 %. This consistency confirms the robustness of the ANN in capturing the rapid and nonlinear thermal dynamics of the pond surface layer.

In the open cycle, T_{C1} increases from 34.00 °C at 07:00 h to a maximum of 58.38 °C between 13:00 and 14:00 h. The ANN predictions closely follow this behavior with minimal error (MRE = 0.44 %). The rise is governed by the steady increase in solar irradiance, which enhances absorption at the pond surface and elevates the cover

Table 3

The results of the statistical analysis of errors for different types of training functions.

Function	N° of hidden layers	N° of neurons	SSE	RMSE (%)	MRE (%)	R^2	Training Time (s)	Best Epoch
trainbr	1	5	999.9238	48.1107	0.87	0.9999	3	310
		10	645.2234	38.6468	0.72	0.9999	7	440
	2	5	681.3693	39.7145	0.74	0.9999	3	267
		15	463.5746	32.7580	0.62	0.9999	93	870
	3	10	488.4118	33.6241	0.62	0.9999	66	1000
		5	594.4154	37.0940	0.70	0.9999	4	100
trainlm	1	5	1484.7205	58.6247	0.98	0.9998	1	43
	2	10	594.4154	37.0940	0.70	0.9999	4	100
	3	5	634.6896	38.3300	0.71	0.9999	3	213
trainbfg	1	10	1429.7470	57.5291	0.97	0.9998	1	89
	2	5	1020.6553	48.6069	0.85	0.9999	1	146
	3	15	604.7769	37.4159	0.70	0.9999	24	179
trainscg	1	5	4307.5440	99.8557	1.84	0.9995	1	25
	2	10	1908.5109	66.4669	1.16	0.9998	1	154
	3	20	1466.1337	58.2566	1.05	0.9998	1	180

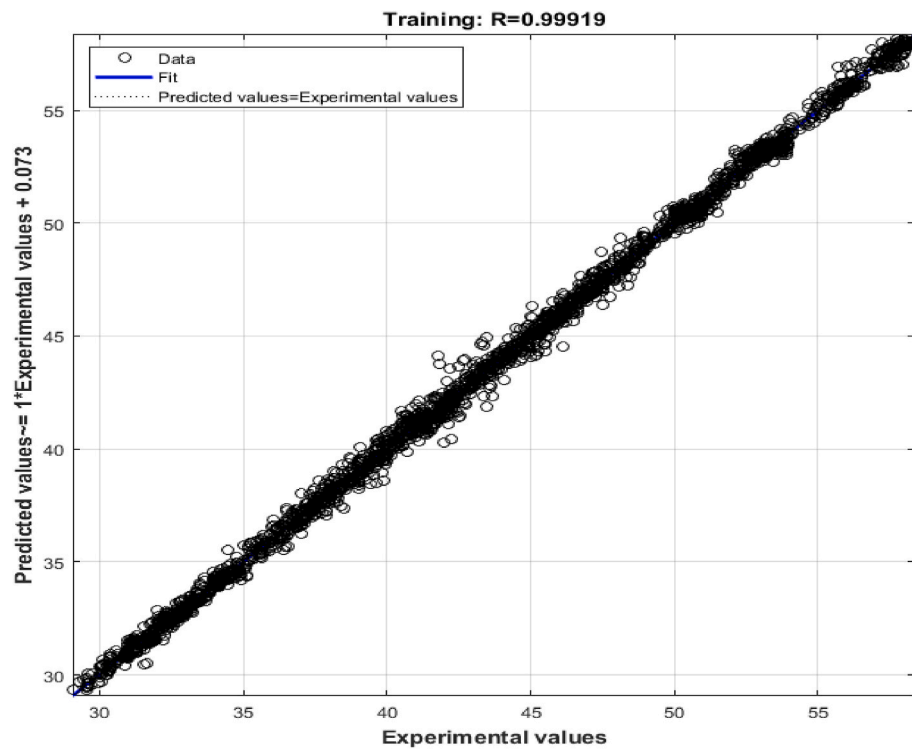


Fig. 6. Comparison of experimental and predicted values during training phase.

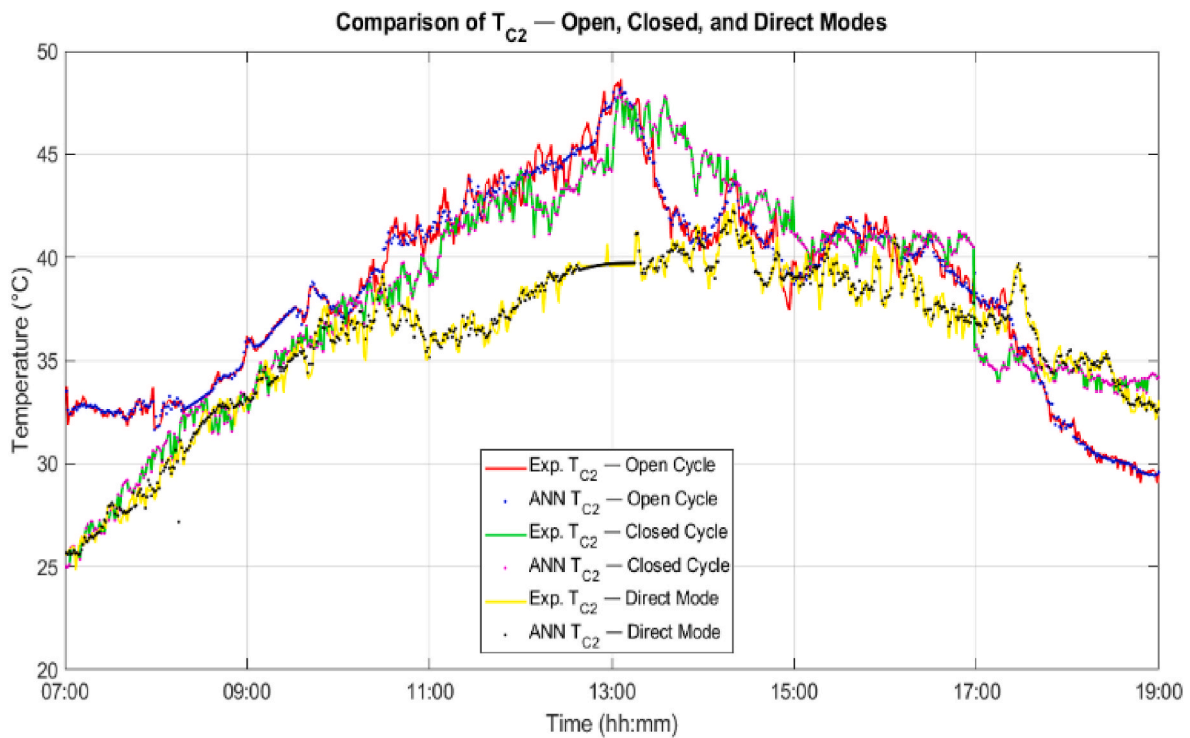


Fig. 7. Experimental vs Predicted temperatures of the upper glass cover in open, closed, and direct modes.

temperature via radiative and convective coupling. The stabilization after midday indicates a balance between incoming solar flux and heat dissipation through convection and longwave radiation. The relatively lower peak compared to other modes reflects stronger convective interactions with ambient air due to the direct replacement of hot pond water with colder fluid, which intensifies cooling losses.

For the closed cycle, the temperature starts at 25.00 °C and gradually climbs to 63.00 °C, peaking later in the afternoon (around 15:20). ANN predictions remain closely aligned, with small discrepancies at peak hours (MRE = 0.48 %). The delayed peak compared to the open cycle suggests that the closed-loop exchanger reduces early convective interactions with the ambient environment, allowing a more gradual

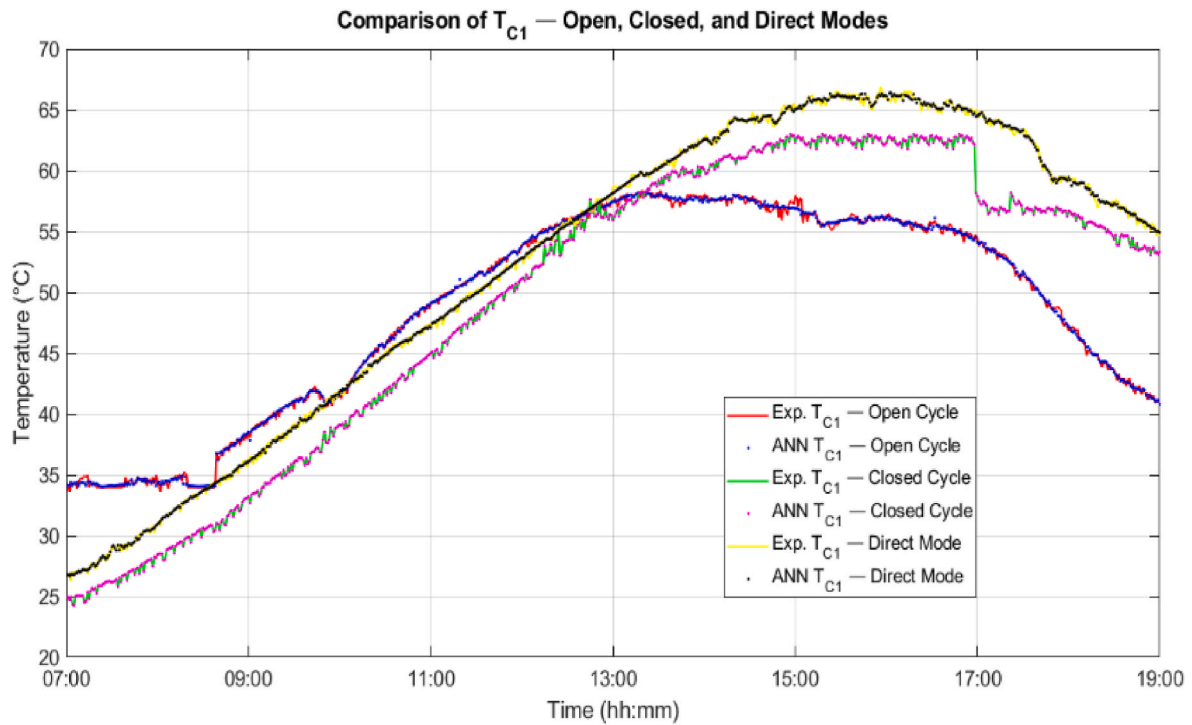


Fig. 8. A comparison between measured and simulated lower glass cover temperatures for different heat extraction modes.

thermal build-up. The higher maximum compared to the open cycle confirms the improved insulation effect of this configuration, as evaporative and convective losses to the surroundings are minimized. This enhances both retention and stability, which are advantageous for sustained thermal utilization.

The direct mode demonstrates the broadest range, from 26.75 °C in the morning to the highest observed peak of 66.87 °C at 16:00 h. The ANN model successfully tracks this trend, though it slightly

underestimates post-peak cooling (MRE = 0.42 %). The sharp rise in this mode is explained by the absence of a thermal exchange barrier, which allows rapid and direct transfer of solar heat to the pond surface. However, this same mechanism promotes strong convective exchanges with ambient air, resulting in large diurnal swings. The underestimation of ANN predictions during the afternoon is attributed to nonlinear radiative losses—particularly longwave radiation to the cooler sky—and enhanced convective heat transfer, which are challenging to reproduce

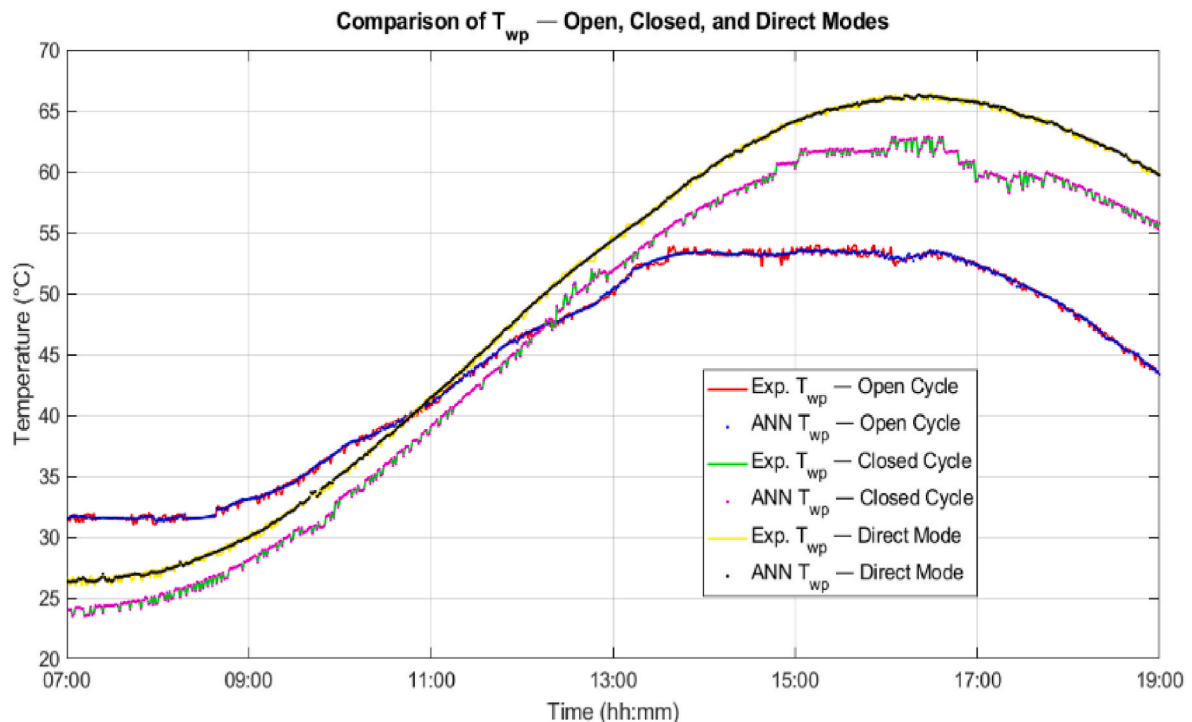


Fig. 9. Pond water temperature: experimental data and ANN predictions across open, closed, and direct cycles.

perfectly.

Fig. 9 presents a comparative evaluation of the experimental and ANN-predicted pond water temperature (T_{wp}) across the three heat extraction modes: open, closed, and direct. As T_{wp} represents the effective thermal output of the pond closely tied to the quality and usability of extracted energy its accurate prediction is of high practical importance.

In the open cycle, T_{wp} rises from 32 °C in the morning to a plateau of 53.94 °C between 13:00 and 15:00 h. ANN estimates follow the experimental curve with remarkable precision (maximum deviation $\approx$ 0.39 °C, MRE = 0.46 %). The midday stabilization reflects a transient equilibrium between the rising solar irradiance and simultaneous convective and radiative losses at the air–water interface. The sudden reductions observed during withdrawal events confirm the inherently unstable nature of this mode, as the direct replacement of hot pond water with colder fluid induces sharp cooling. Nevertheless, the ANN model successfully reproduces these fluctuations, validating its capability to track rapid thermal changes.

In the closed cycle, the water temperature begins at 25 °C and steadily climbs to a maximum of 62.88 °C around 16:00 h. ANN predictions remain tightly aligned, with minor underestimations during the cooling phase (MRE = 0.54 %). The delayed yet higher peak relative to the open cycle highlights the insulating effect of the closed-loop exchanger, which minimizes evaporative and convective losses. This configuration enables deeper and more gradual accumulation of thermal energy within the pond, resulting in greater thermal stability and higher output quality. The slight deviations between ANN and experiment during the evening cooling period are linked to nonlinear exchanges particularly sky radiation and convective currents which are challenging to model with complete accuracy.

The direct mode demonstrates the widest range, from 27 °C at 07:00 h to a maximum of 66.44 °C at 16:00 h. ANN predictions capture this strong upward trend with high fidelity (MRE = 0.47 %), though slight underestimations occur post-peak. The rapid heating arises from the

absence of an insulating exchanger, which allows immediate and direct energy accumulation in the pond water. However, this also exposes the system to stronger convective coupling with the ambient environment, leading to larger diurnal fluctuations. The post-peak discrepancies between ANN and experimental data reflect intensified radiative losses to the cooler sky and convective cooling to ambient air.

Fig. 10 illustrates a comparative analysis between experimental and ANN-predicted absorber plate temperatures (T_p) under the three heat extraction modes: open, closed, and direct. The absorber plate, located at the pond base, plays a critical role in capturing and redistributing solar flux into the shallow water mass, making its thermal behavior central to overall system efficiency.

In the open cycle, T_p rises from 33 °C at 07:00 h to a maximum of 53.94 °C between 13:00 and 15:00 h. ANN predictions maintain close agreement, with only slight deviations (maximum $\approx$ 0.13 °C, MRE = 0.50 %). The relatively lower peak temperature compared to the other modes results from continuous withdrawal of hot water, which enhances convective cooling at the absorber surface and promotes evaporative losses at the air–water interface. These effects combine to reduce the overall retention of absorbed heat, though the ANN model accurately reproduces both the heating and cooling trajectories.

The closed cycle begins at 26 °C and gradually increases to 62.02 °C by 16:00 h. ANN predictions closely follow this progression, with only minor underestimations during the late cooling phase (MRE = 0.52 %). The higher peak relative to the open cycle reflects reduced evaporative and convective losses due to the isolated heat exchanger configuration. In this mode, the absorber plate effectively channels absorbed solar flux into the bulk water, where heat accumulates more stably and with less disruption from ambient air interactions.

The direct mode shows the steepest rise, from 28 °C in the morning to 66.36 °C at 16:00 h. The ANN model captures this rapid heating trend with excellent accuracy (MRE = 0.42 %), though slight post-peak variations are observed. The absence of an exchanger barrier allows for maximum heat buildup, but also increases susceptibility to strong

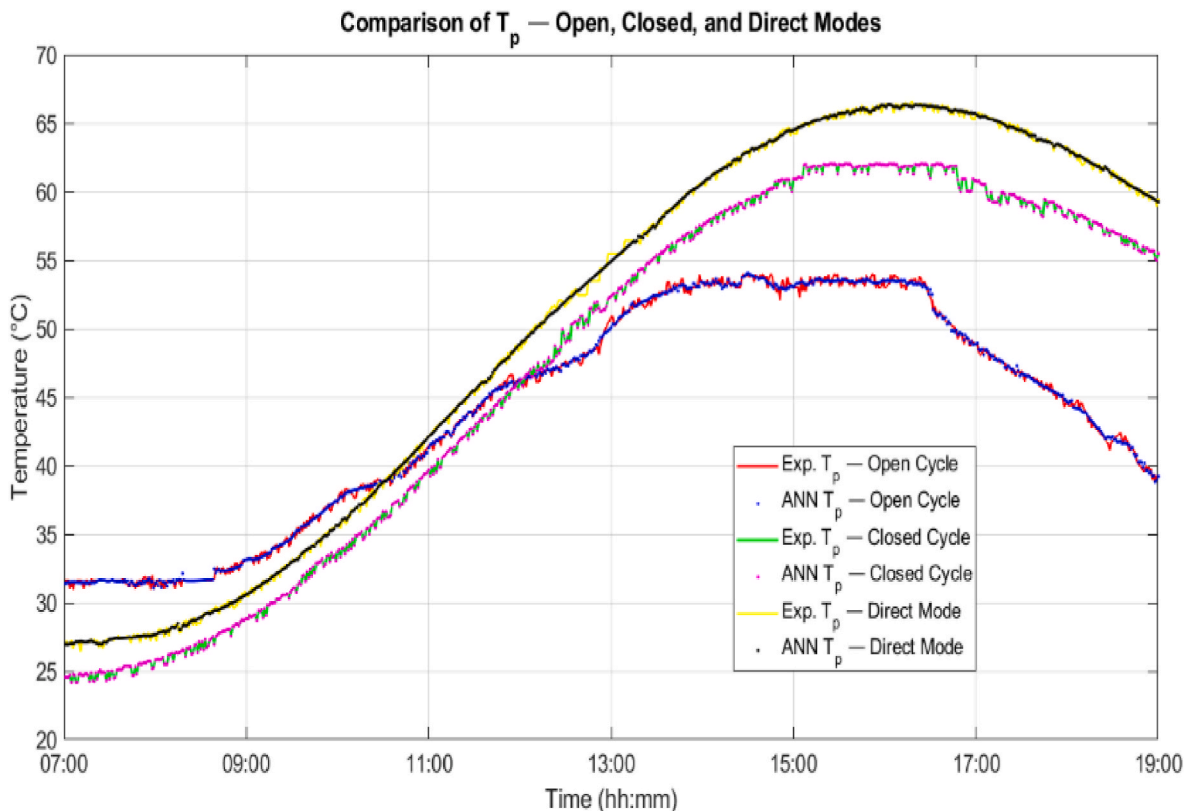


Fig. 10. Comparison of experimental and predicted absorber plate temperatures for open, closed, and direct modes.

ambient coupling, explaining the sharp diurnal range. The post-peak deviations are attributable to enhanced radiative losses from the absorber to the sky and intensified convective cooling during evening hours.

An important observation from Fig. 10 is the closeness between absorber plate, pond water, and lower glass cover temperatures, a direct consequence of the shallow pond geometry. The minimal water depth leads to weak vertical gradients, effectively homogenizing temperatures across components. The lower glass cover aligns closely with bulk water due to direct thermal contact, though it often registers slightly higher values from direct solar irradiation and reduced dissipation to the exchanger. Similarly, the absorber plate, positioned at the pond base, absorbs direct solar flux that is quickly redistributed through conduction and convection in the shallow water volume. This strong thermal coupling underscores the highly interactive nature of shallow solar ponds, a dynamic faithfully reproduced by the ANN model.

Fig. 11 presents the experimental and ANN-predicted outlet fluid temperature (T_{fo}) profiles for the open and closed cycles, noting that the direct mode is absent since T_{fo} is only relevant under active heat extraction. The results show that the ANN model reproduces the experimental trends with very high accuracy in both configurations. In the open cycle, the outlet temperature rises from 30 °C at 07:00 h and stabilizes around 51.5 °C between 13:00–16:00 h, with only minor deviations from experimental values (max ≈ 0.29 °C, MRE = 0.66 %). In the closed cycle, the fluid warms from 25 °C to a delayed but higher peak of 58.88 °C at 16:00 h, with the ANN again matching closely, except for slight underestimations during the cooling phase (MRE = 0.63 %). These patterns highlight the different thermal mechanisms of each configuration: the open cycle is constrained by direct heat transfer fluid replacement, convective cooling, and enhanced evaporative losses, which accelerate stabilization of the outlet temperature around midday; while the closed cycle benefits from exchanger isolation, allowing deeper accumulation of heat and smoother delivery, which explains its higher and more stable outlet temperature. Comparatively, the open

cycle favors faster but less stable extraction, while the closed cycle ensures higher-quality thermal energy with better buffering capacity and reduced sensitivity to ambient fluctuations. Since T_{fo} represents the final useable energy of the system, its sensitivity to extraction strategy makes it a critical indicator of system efficiency.

Fig. 12 presents the experimental and ANN-predicted water tank temperature (T_{wt}) profiles under the closed-cycle configuration. Starting from 25 °C at 07:00 h, the tank temperature increases steadily throughout the day, reaching a maximum of 53.56 °C around 17:00 h before stabilizing. The ANN model captures this thermal evolution with high accuracy, with only slight deviations during the final stabilization phase (MRE = 0.68 %). This smooth and continuous rise reflects the role of the insulated storage tank as a thermal buffer, which minimizes the influence of ambient convective and radiative disturbances and ensures gradual accumulation of heat. Unlike the pond water that directly exchanges energy with the environment, the tank retains thermal energy more effectively, producing a delayed but stable temperature profile. The eventual stabilization of T_{wt} indicates that thermal equilibrium is reached when the heat input from the solar pond balances the losses through the tank walls, while the small discrepancies in ANN predictions are likely linked to nonlinear insulation effects and delayed heat response. These results confirm the superior storage capacity and thermal inertia of the closed cycle, which enhances the system's ability to preserve heat and deliver stable thermal energy for extended use. The close agreement between ANN predictions and experiments further demonstrates the robustness of the model, supporting its potential application not only in forecasting but also in real-time optimization of thermal storage management within solar pond systems.

Fig. 13 illustrates the experimental temperature variations of the solar pond under the three extraction modes, alongside solar radiation intensity, which peaked at 909.82 W/m² at 12:00 h. A clear thermal lag is observed between the radiation maximum and the peak fluid temperatures, a result of the water's high thermal inertia and slow heat transfer dynamics. This effect is most pronounced in the closed cycle and

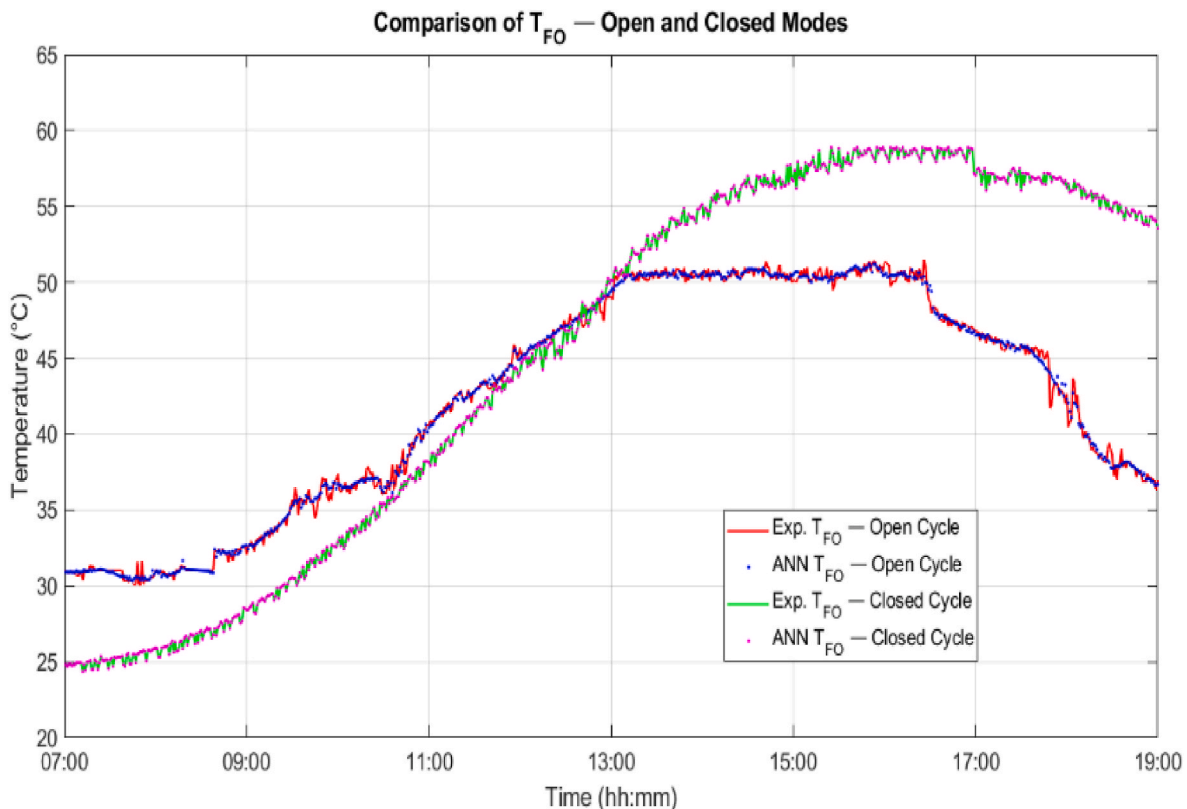


Fig. 11. Measured and simulated heat transfer fluid temperature profiles: observations vs. ANN predictions.

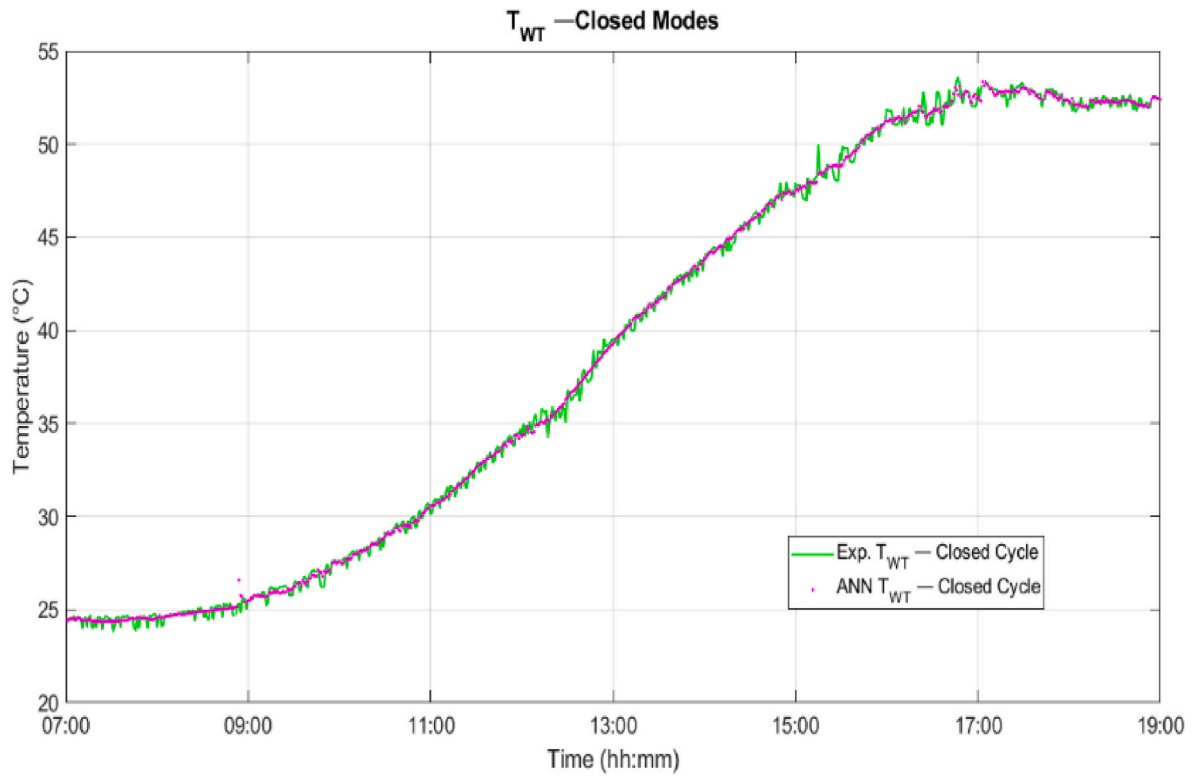


Fig. 12. Storage tank temperature: experimental results versus model estimates.

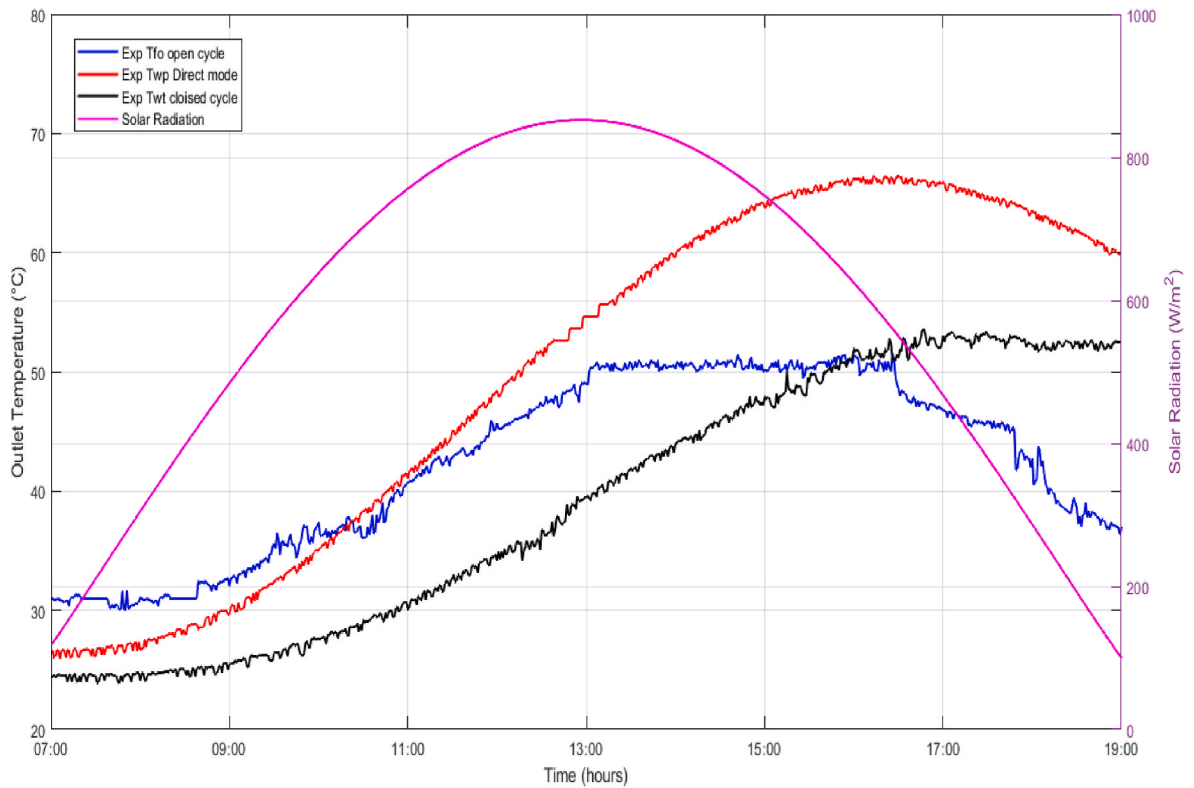


Fig. 13. Comparison of experimental fluid and water temperatures for open, closed, and direct modes with solar radiation.

direct extraction modes, where the larger thermal mass delays but sustains the heating response. By contrast, the open-cycle mode exhibits a faster response to solar fluctuations due to continuous circulation and

direct mixing, though this results in lower peak values and reduced storage capacity. The closed cycle achieves the most stable and extended thermal retention, maintaining elevated temperatures well into the

evening and thereby minimizing losses, while the direct mode produces higher peaks but with stronger fluctuations and rapid cooling once solar input diminishes. The open cycle, although more moderate, provides a relatively balanced performance but with lower long-term efficiency. These results are consistent with earlier findings by Aboul-Enein et al. [24], Ramadan et al. [25], El-Sebaei et al. [26], and El-Sebaei et al. [27], confirming the superior energy conservation capability of the closed configuration.

7. Model predictive control design and parameter identification

Building on the experimental comparison of the three extraction modes (Fig. 13), it was shown that the closed cycle delivers the most stable and extended heat retention, ensuring minimal thermal losses and sustained elevated temperatures. This superior stability makes the closed configuration the most suitable candidate for advanced control enhancement. Accordingly, this section develops a hybrid ANN-MPC strategy aimed at regulating the fluid flow rate in the SSP closed cycle. The controller is designed with two main objectives: (i) minimizing the outlet temperature tracking error, and (ii) reducing the energy consumption of the circulation pump. Implementation is carried out on the QR30E brushless DC pump, which operates within a constrained flow range, making it particularly well-suited for predictive optimization guided by the ANN model.

7.1. Controller concept

The MPC controller estimates the optimal flow rate by simulating the future outlet temperature response for a set of candidate flow rates. For each candidate value, the temperature correction is evaluated using a simplified dynamic model:

$$T_{fo,corr} = T_{fo,pred} + \alpha \cdot \tanh(\beta(\dot{m} - 0.01)) \quad (5)$$

where:

1. $T_{fo,corr}$: the corrected or future temperature after applying the MPC adjustment.
2. $T_{fo,pred}$: the temperature predicted by the ANN model without correction.
3. $\dot{m}$: the mass flow rate of the working fluid in the heat exchanger (kg/s).
4. α : the gain factor, representing the magnitude of the thermal correction applied by the MPC.
5. β : the sensitivity factor, which controls how sharply the tanh function responds to variations in the flow rate.
6. 0.010: the reference set-point for the mass flow rate, approximately corresponding to the minimum operational flow rate of the QR30E pump used in the system.

The overall workflow of the proposed ANN-MPC control strategy, including the prediction, decision, and feedback loop, is illustrated in Fig. 14.

7.1.1. Determination of α and β parameters

To effectively tune the MPC, it is essential to determine the correction parameters α and β . These parameters were identified through an offline optimization procedure, in which the objective was to minimize the error between the desired outlet temperature (T_{fo-set}) and the ANN-predicted temperature ($T_{fo,pred}$). The experimental mass flow rate profile was employed as input during this optimization, ensuring that the identified parameters reflect realistic operating conditions. The optimization was carried out using the fminsearch algorithm, which iteratively adjusted α and β to reduce the mean squared error (MSE) between the target and predicted temperatures.

The obtained results, illustrated in Fig. 15, confirm that the corrected

temperature trajectory closely follows the setpoint. This outcome validates the suitability of the optimized parameters ($\alpha = -22.93619$, $\beta = 33.23245$) for incorporation into the MPC framework, as they encapsulate the system's dynamic behavior. The corrected output ($T_{fo-fmin-search}$) achieved a low MSE of 0.98907, demonstrating the effectiveness of this approach for enhancing prediction accuracy and ensuring robust real-time control performance.

7.2. Neural Network-Based MPC control system architecture

Fig. 16 illustrates the architecture of an MPC system integrated with an ANN for effective temperature regulation. This hybrid system combines the predictive capabilities of an ANN with the optimization potential of MPC to ensure precise control over the system's temperature.

7.2.1. System overview

The architecture consists of the following key components:

- Input: The initial temperature and process input to the system.
- Function Fitting Neural Network (FNN): A data-driven model trained to predict the system's output temperature $T_{fo-corr}$ based on input dynamics and historical data.
- MPC Controller: Utilizes the predictions from the ANN to optimize the flow rate in order to minimize the error between the $T_{fo-pred}$ and the T_{fo-set} .
- Error Signal: The difference between the $T_{fo-pred}$ and the T_{fo-set} serves as the primary input to the MPC, driving corrective actions to adjust the flow rate and maintain temperature stability.

7.2.2. System workflow

The system operates in a feedback loop, and its workflow can be broken down into the following steps:

1. Neural Network Prediction:

The FNN processes the input data and simulates the expected output temperature $T_{fo-corr}$. The ANN's predictive model captures the system's dynamics, providing an estimation of the future temperature response.

2. MPC Optimization:

The MPC controller evaluates the predicted temperature against the desired setpoint.

Using the predefined parameters α and β (discussed in Section 7.1), the MPC calculates the optimal mass flow rate. This optimization minimizes the error between the predicted temperature and the setpoint, ensuring efficient system operation.

3. Closed-Loop Correction:

Based on the optimization by the MPC, the adjusted flow rate is applied to the system. The process is repeated in real-time, continuously correcting the flow rate to ensure temperature stability and to meet the target setpoint.

Fig. 17 illustrates a comparison between mass flow rate patterns observed under two different conditions: the first controlled by an MPC, and the second recorded during the training phase of the ANN.

The mass flow rate managed by the MPC exhibits a more adaptive and energy-efficient pattern. Starting at approximately 9×10^{-3} kg/s in the early morning, the flow rate gradually decreases and remains relatively low during midday hours when the temperature naturally rises. The system then increases the flow rate in the late afternoon as solar radiation declines.

This behavior confirms that the MPC effectively and dynamically adjusts the mass flow rate in real time to minimize thermal losses while maintaining a temperature close to the desired setpoint.

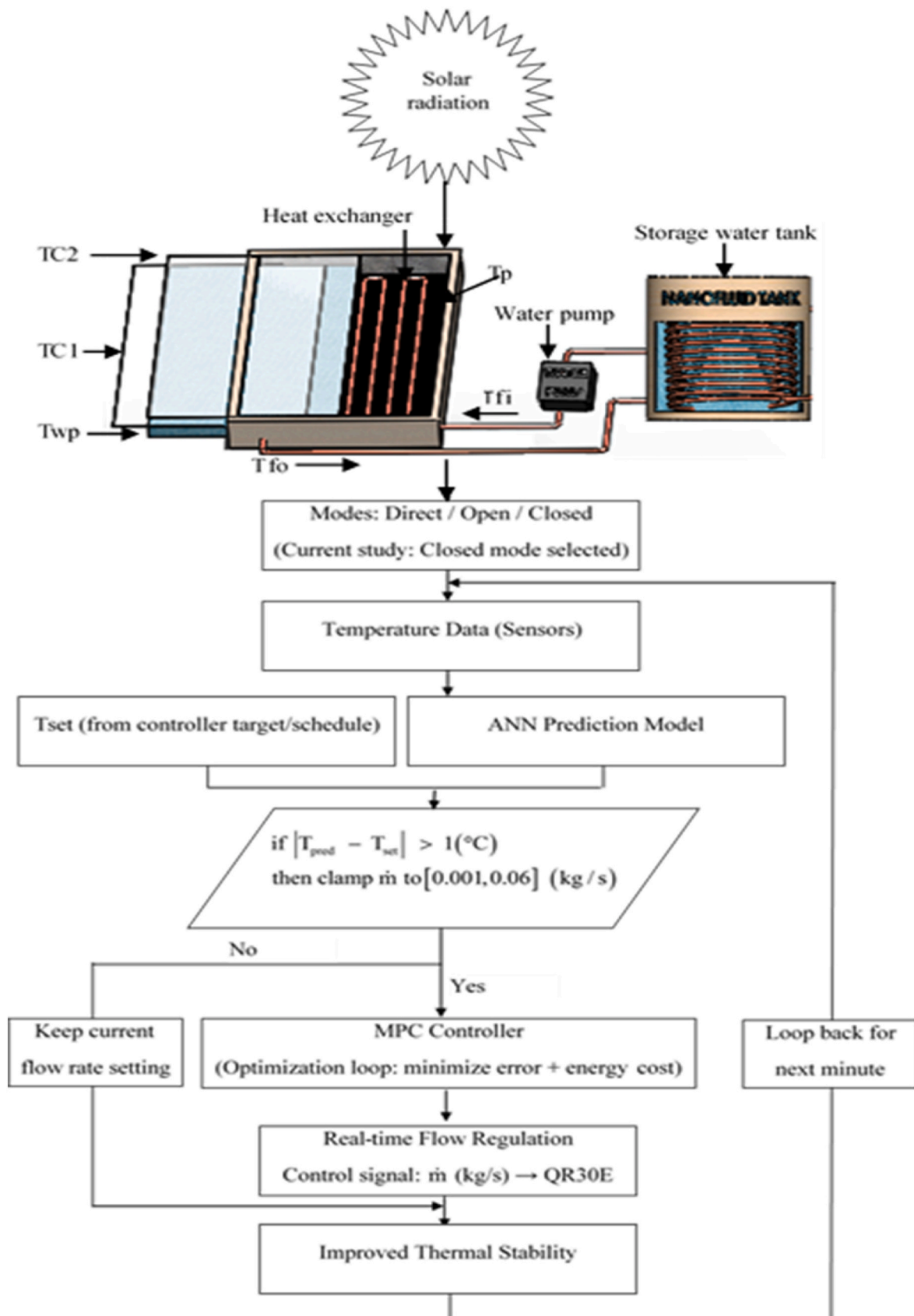


Fig. 14. Integrated flow process and ANN-MPC control loop for real-time thermal regulation in the SSP.

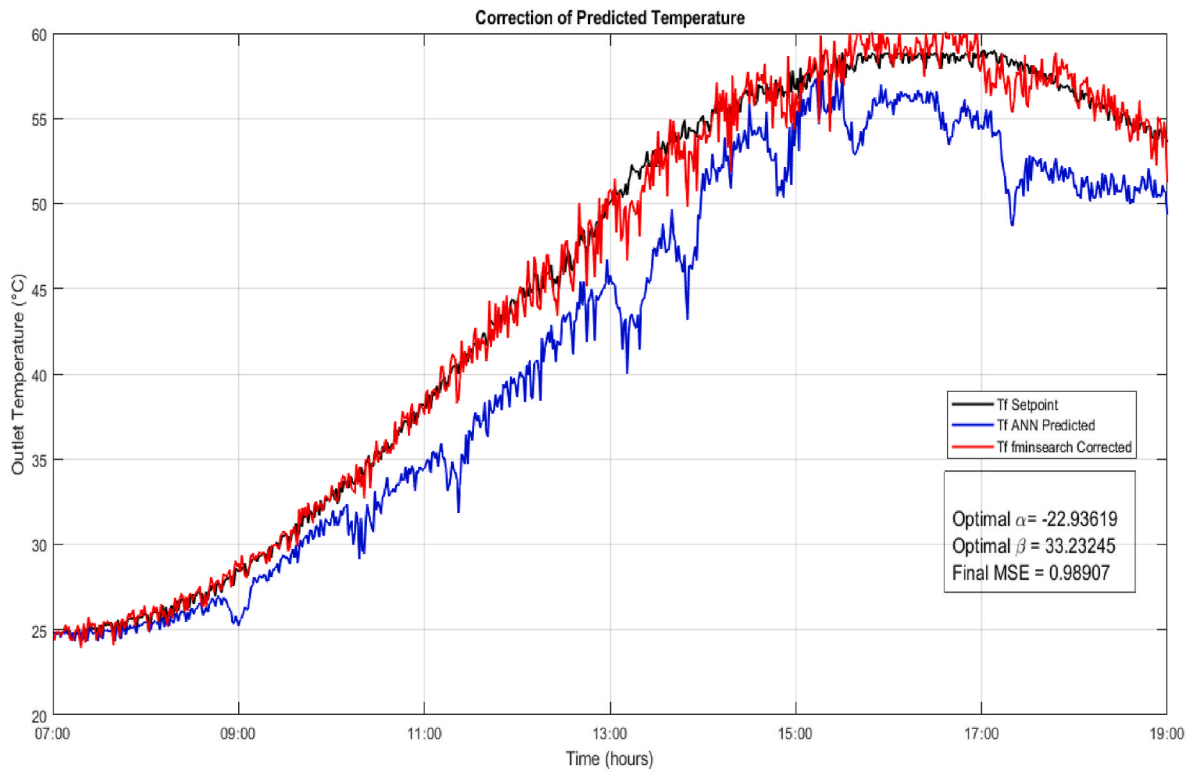


Fig. 15. Optimization of Parameters α and β for Temperature Prediction Correction.

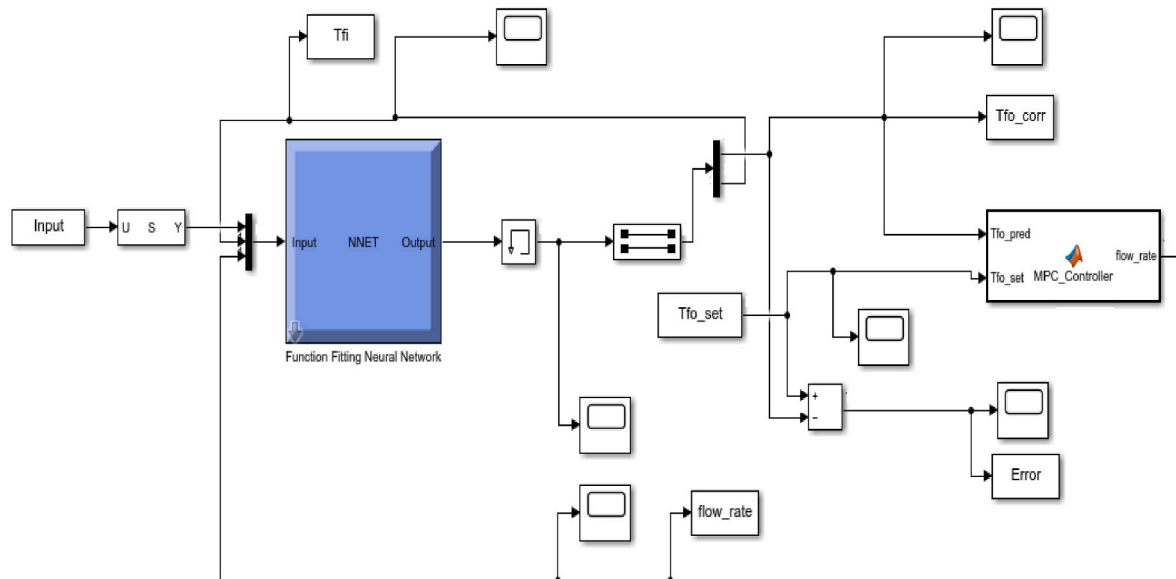


Fig. 16. Schematic of a neural network-based MPC system for temperature control.

Fig. 18 presents a comparative analysis of the outlet temperature profiles under three distinct operational scenarios. The red curve, representing the outlet temperature corrected by the MPC system $T_{fo-corr}$, clearly exceeds in performance the black curve, which corresponds to the temperature predicted by the ANN without any corrective action $T_{fo-pred}$. Although both temperature profiles were obtained under the same reduced solar irradiance conditions (green curve, recorded on November 17), the red curve consistently remains in closer proximity to the reference temperature T_{fo-set} (blue curve), particularly during the critical midday and afternoon hours. This improved performance is attributed to the adaptive regulation of the mass flow rate provided by

the MPC system, which adjusts the fluid flow in real time. Consequently, the working fluid resides for a longer duration within the heat exchanger during periods of low solar irradiance, enabling more efficient heat capture and reducing thermal losses.

Quantitatively, the maximum outlet temperature achieved by the MPC system was approximately 52.1 °C, whereas the temperature predicted by the ANN without correction peaked at 49.3 °C, resulting in a 2.8 °C advantage for the MPC approach. The minimum outlet temperatures were 24.9 °C for the corrected temperature profile and 24.0 °C for the predicted profile, indicating enhanced early-morning heat retention. With respect to prediction accuracy, the mean absolute error relative

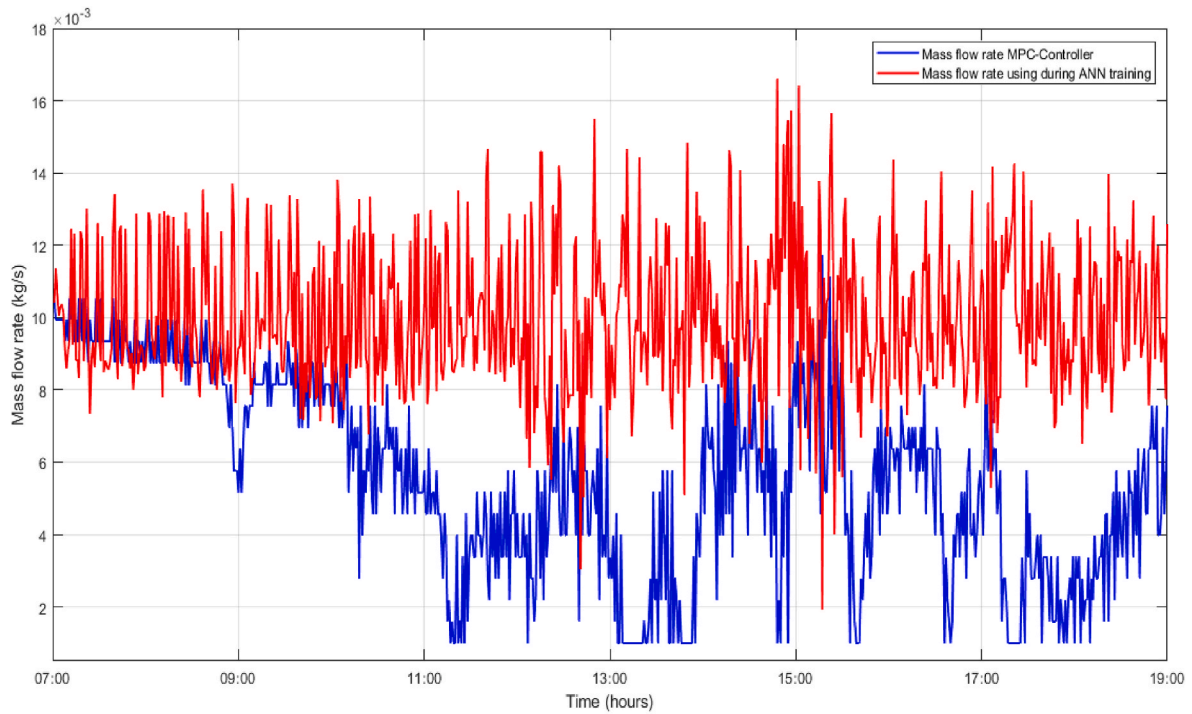


Fig. 17. Comparison of MPC-Controlled vs. ANN-Trained Mass Flow Rates.

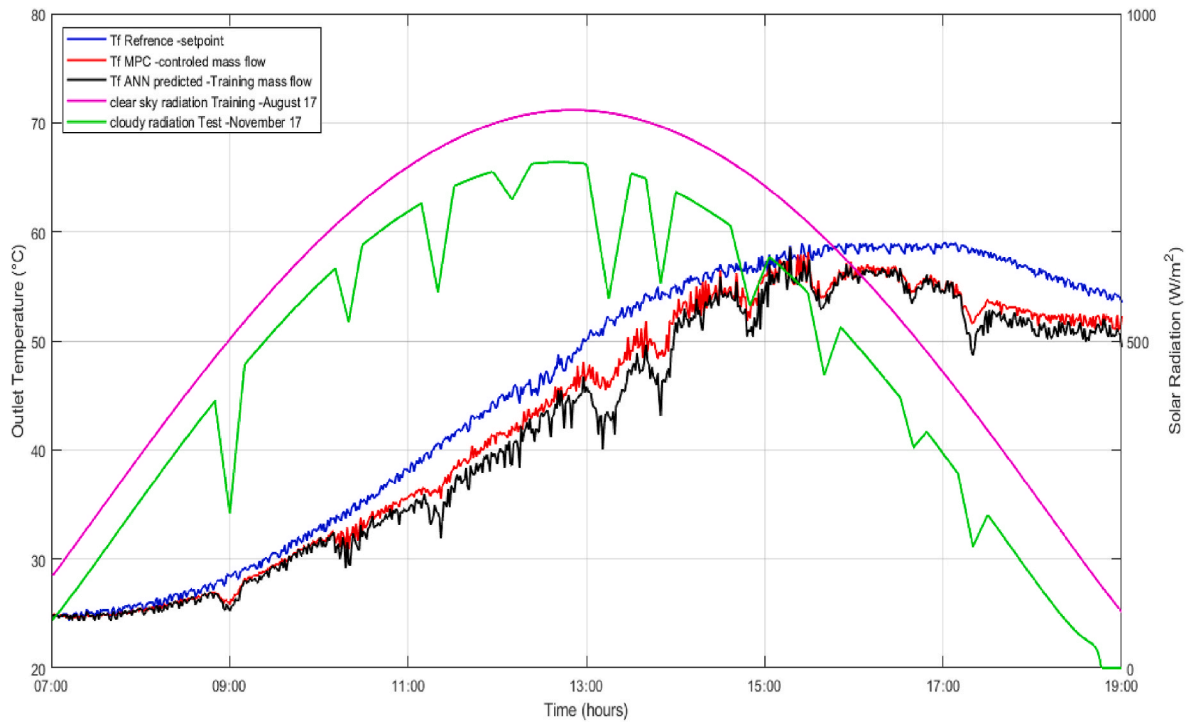


Fig. 18. Outlet Temperature Comparison under ANN-Trained vs. MPC-Controlled Flow Rates with Varying Solar Radiation.

(MAE) to the reference temperature T_{fo-set} was $1.42\text{ }^{\circ}\text{C}$ for the corrected profile, compared to $2.97\text{ }^{\circ}\text{C}$ for the predicted profile, representing a 52.2 % reduction in error. This substantial improvement confirms the superior tracking capability of the MPC system and its effectiveness in maintaining temperature stability through intelligent adjustment of the mass flow rate.

Both the corrected and predicted temperature profiles display distinct dips throughout the day, directly corresponding to fluctuations

in solar irradiance as observed in the green curve, particularly between 09:30 and 16:00, when radiation dropped below 500 W/m^2 . These variations reflect the system's thermal response to cloudy weather conditions. In contrast, the reference temperature profile T_{fo-set} , derived from ANN predictions using clear-sky solar data from August 17 (purple curve), exhibits a smooth and uninterrupted temperature trajectory, with solar irradiance peaking around 950 W/m^2 and remaining stable.

This comparative analysis highlights the sensitivity of the SSP to

variations in solar radiation and emphasizes the effectiveness of the MPC system in mitigating these challenges. By continuously adjusting the mass flow rate in response to real-time environmental inputs, the system enhances thermal regulation, improves adherence to the desired outlet temperature, and maintains high operational efficiency even under non-ideal solar conditions.

8. Conclusion

This study successfully demonstrated the superior thermal performance of a closed-cycle SSP, insulated and equipped with double-glazing, a PVC heat exchanger, and a black-painted bottom, and the efficacy of a hybrid ANN and MPC framework for its optimization. Experimental results under identical environmental conditions confirmed that the closed-cycle mode outperformed both direct and open-cycle modes, providing higher peak temperatures, enhanced thermal stability, and prolonged heat retention due to minimized convective and evaporative losses.

The developed ANN model proved to be a powerful tool for predicting the complex, nonlinear thermal behavior of the SSP. The optimally configured network (trainbr algorithm, 2 hidden layers, 15 neurons) achieved remarkable accuracy across all measured temperatures (T_{C1} , T_{C2} , T_{wp} , T_p , T_{fo} , T_{wt}) and under all operational modes, validating its capability to handle both steady-state and dynamic conditions.

The most significant contribution of this work is the integration of this accurate ANN model into a real-time MPC strategy. By dynamically adjusting the mass flow rate based on predictive control, the MPC-ANN system reduced thermal tracking errors by more than 50 %, significantly enhancing the system's efficiency and robustness against solar irradiance variations. The optimized control parameters ($\alpha = -22.94$, $\beta = 33.23$) for the closed-cycle operation highlight the potential for adaptive, energy-efficient management of solar thermal systems.

In summary, this research provides:

1. A definitive experimental comparison establishing the closed-cycle configuration as the most efficient method for heat extraction from an SSP.
2. A highly accurate, data-driven ANN model for the thermal simulation of SSPs.
3. A practical and intelligent MPC-ANN control framework that significantly improves real-time operational efficiency and thermal output stability.

This integrated approach paves the way for the intelligent automation and optimization of renewable energy systems, promising greater reliability and efficiency for practical, large-scale solar thermal applications. Future work will focus on extending this strategy to multi-objective optimization and real-world hardware-in-the-loop validation.

CRediT authorship contribution statement

Abdelkrim Terfai: Writing – original draft, Validation, Software, Resources, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Younes Chiba:** Writing – review & editing, Visualization, Validation, Supervision, Investigation, Data curation, Conceptualization. **Mounir Zirari:** Writing – review & editing, Validation, Supervision, Project administration. **Mohamed Najib Bouaziz:** Writing – review & editing, Validation, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research was supported by the Renewable Energy and Materials Laboratory, University Yahia Fares of Medea, Algeria.

Nomenclature

Geometric and physical parameters

- A Area (m^2)
- D_{he} Tube diameter of the pond heat exchanger (m)
- L_{he} Tube length of the pond heat exchanger (m)
- δ_{is} Insulating material thickness (m)
- δ_w Depth of the shallow solar pond (m)
- k Thermal conductivity ($W/m \cdot K$)
- ρ Density (kg/m^3)
- C_p Specific heat capacity ($J/kg \cdot K$)
- m_{wt} Mass of water in the storage tank (kg)
- m Mass flow rate (kg/s)

Thermal and experimental variables

- T_a – Air (ambient) temperature ($^{\circ}C$)
- T_p – Absorber plate temperature ($^{\circ}C$)
- T_{C1} – Lower glass cover temperature ($^{\circ}C$)
- T_{C2} – Upper glass cover temperature ($^{\circ}C$)
- T_{wp} – Pond water temperature ($^{\circ}C$)
- T_{fi} – Inlet fluid temperature to heat exchanger ($^{\circ}C$)
- T_{fo} – Outlet fluid temperature from heat exchanger ($^{\circ}C$)
- T_{wt} – Storage tank water temperature ($^{\circ}C$)
- T_{fo-set} – Setpoint outlet fluid temperature ($^{\circ}C$)
- $T_{fo-corr}$ – Corrected outlet fluid temperature ($^{\circ}C$)
- $T_{fo-pred}$ – Predicted outlet fluid temperature by ANN ($^{\circ}C$)

Environmental parameters

- I_T – Solar radiation (W/m^2)
- Hum – Relative humidity (%)
- W_{speed} – Wind speed (m/s)
- $Time$ – Desired time period (min)

Heat transfer & optical properties

- Q – Heat transfer rate (W)
- α_p – Absorptivity of absorber plate (–)
- α_g – Absorptivity of glass cover (–)
- τ_g – Transmissivity of glass cover (–)
- ϵ_g – Emissivity of glass cover (–)
- ϵ – Heat exchanger effectiveness (–)

ANN / MPC modeling parameters

- e_i Experimental value
- p_i Predicted value
- n Number of independent data samples
- α, β Regression/ANN model parameters
- ANNs Artificial Neural Networks
- MPC Model Predictive Control
- SSP Shallow Solar Pond

Error analysis & statistics

- SSE – Sum of Squared Errors
- $RMSE$ – Root Mean Square Error
- MRE – Mean Relative Error
- MAE – Mean Absolute Error Relative
- R^2 – Coefficient of determination

References

- [1] K.H. Solangi, M.R. Islam, R. Saidur, N.A. Rahim, H. Fayaz, A review on global solar energy policy, *Renew. Sustain. Energy Rev.* 15 (4) (2011) 2149–2163.

- [2] H.P. Garg, A. Shukla, A.K. Bhargava, Shallow solar pond: state-of-the-art, *Energy Convers. Manag.* 22 (2) (1982) 117–131.
- [3] W.C. Dickinson, J.H. Brown, The shallow solar pond energy conversion system, *Sol. Energy* 18 (1) (1976) 3–10.
- [4] A. Kudish, Sede Boqer shallow pond project, *Energy* 6 (3) (1981) 277–292.
- [5] M.S. Sodha, G.N. Tiwari, J.K. Nayak, Shallow solar pond water heater: an analytical study, *Energy Convers. Manag.* 21 (2) (1981) 137–139.
- [6] P.K. Bansal, D. Buddhi, R.L. Sawhney, Effect of heat exchanger on the performance of a shallow solar pond water heater, *Energy Convers. Manag.* 24 (4) (1984) 259–263.
- [7] A. Terfai, et al., Thermal performance enhancement of a shallow solar pond based on nanofluids, *J. Nano- Electron. Phys.* 12 (1) (2020) 01015, 1–01015-6.
- [8] H.M. Ali, M. Akhlaghi, Experimental study in a shallow solar pond, *Sol. Wind Technol.* 4 (4) (1987) 425–430.
- [9] J. Gonzalez, L.R. Perez, J. Benitez, Modeling the thermal process in a shallow solar pond water heater, *Sol. Energy* 48 (4) (1992) 261–265.
- [10] V. Velmurugan, K. Srithar, Solar stills integrated with a mini solar pond—analytical simulation and experimental validation, *Desalination* 216 (1–3) (2007) 232–241.
- [11] A.A. El-Sebaili, M.R.I. Ramadan, S. Aboul-Enein, N. Salem, Thermal performance of a single-basin solar still integrated with a shallow solar pond, *Energy Convers. Manag.* 49 (10) (2008) 2839–2848.
- [12] A.A. El-Sebaili, S. Aboul-Enein, M.R.I. Ramadan, A.M. Khallaf, Thermal performance of an active single basin solar still (ASBS) coupled to shallow solar pond (SSP), *Desalination* 280 (1–3) (2011) 183–190.
- [13] T. Khatib, A. Mohamed, K. Sopian, A review of solar energy modeling techniques, *Renew. Sustain. Energy Rev.* 16 (5) (2012) 2864–2869.
- [14] H.K. Ghritlahre, R.K. Prasad, Application of ANN technique to predict the performance of solar collector systems—A review, *Renew. Sustain. Energy Rev.* 84 (2018) 75–88.
- [15] E.D. Reyes-Téllez, A. Parrales, G.E. Ramírez-Ramos, J.A. Hernández, G. Urquiza, M.I. Heredia, F.Z. Sierra, Analysis of transfer functions and normalizations in an ANN model that predicts the transport of energy in a parabolic trough solar collector, *Desalination Water Treat.* 200 (2020) 23–41.
- [16] H. Benli, Determination of thermal performance calculation of two different types of solar air collectors with the use of artificial neural networks, *Int. J. Heat Mass Tran.* 60 (2013) 1–7.
- [17] A. Ahmad, H.K. Ghritlahre, P. Chandrakar, Implementation of ANN technique for performance prediction of solar thermal systems: a comprehensive review, *Trends Renew. Energy* 6 (1) (2020) 12–36.
- [18] G. Sadeghi, S. Nazari, M. Ameri, F. Shama, Energy and exergy evaluation of the evacuated tube solar collector using Cu₂O/water nanofluid utilizing ANN methods, *Sustain. Energy Technol. Assessments* 37 (2020) 100578.
- [19] A. Abidi, A.I. Khair, R. Kalbasi, Using ANN techniques to forecast thermal performance of a vacuum tube solar collector filled with SiO₂/EG-water nanofluid, *J. Taiwan Inst. Chem. Eng.* 128 (2021) 301–313.
- [20] A.H. Elsheikh, et al., Modeling of solar energy systems using artificial neural network: a comprehensive review, *Sol. Energy* 180 (2019) 622–639.
- [21] K. Kamiuto, T. Oda, Thermal performance of a shallow solar-pond water heater with semitransparent, multilayer surface insulation, *Energy* 16 (10) (1991) 1239–1245.
- [22] R.E. Bird, R.L. Hulstrom, Simplified Clear Sky Model for Direct and Diffuse Insolation on Horizontal Surfaces, Solar Energy Research Institute, Golden, CO, USA, 1981. Technical Report SERI/TR-642-761.
- [23] S. Haykin, *Neural Networks and Learning Machines*, third ed., Pearson, Upper Saddle River, NJ, USA, 2009.
- [24] S. Aboul-Enein, A.A. El-Sebaili, M.R.I. Ramadan, A.M. Khallaf, Parametric study of a shallow solar pond under the batch mode of heat extraction, *Appl. Energy* 78 (2) (2004) 159–177.
- [25] M.R.I. Ramadan, A.A. El-Sebaili, S. Aboul-Enein, A.M. Khallaf, Experimental testing of a shallow solar pond with continuous heat extraction, *Energy Build.* 36 (9) (2004) 955–964.
- [26] A.A. El-Sebaili, S. Aboul-Enein, M.R.I. Ramadan, A.M. Khallaf, Thermal performance of shallow solar pond under open cycle continuous flow heating mode for heat extraction, *Energy Convers. Manag.* 47 (7–8) (2006) 1014–1031.
- [27] A.A. El-Sebaili, S. Aboul-Enein, M.R.I. Ramadan, A.M. Khallaf, Thermal performance of shallow solar pond under open and closed cycle modes of heat extraction, *Sol. Energy* 95 (2013) 30–41.